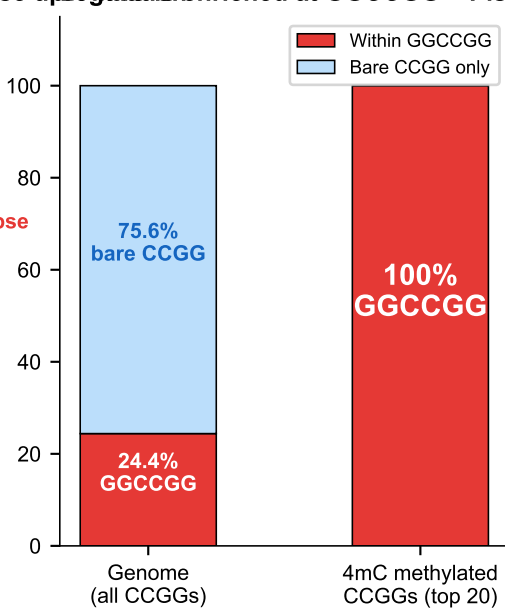
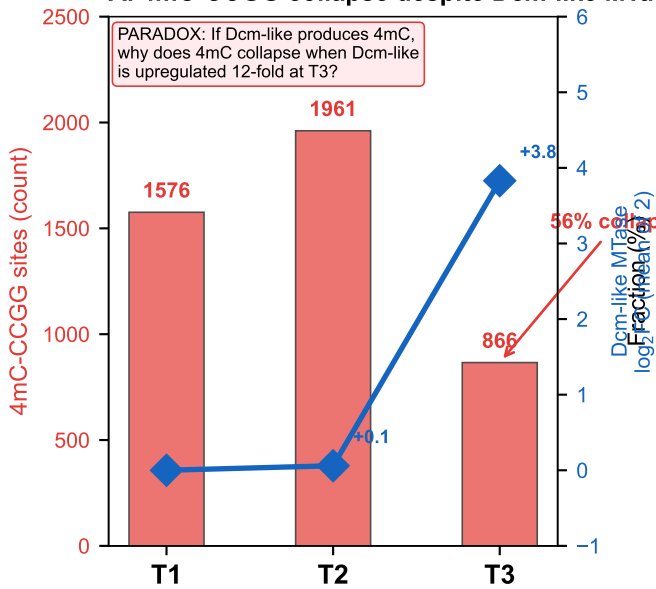
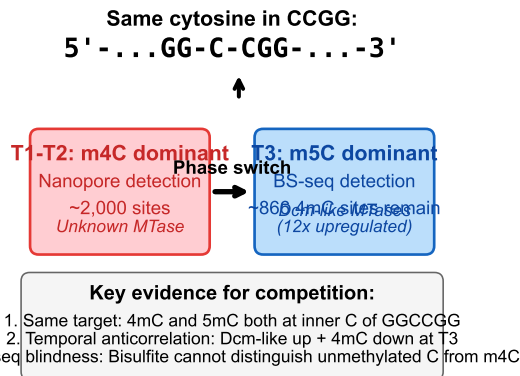


A. 4mC-CCGG collapse despite Dcm-like MTase upregulation



Pisciotta 2023 BS-seq: 5mC at GGCC^{m5}CGG
This study Nanopore: 4mC at GGCC^{m4}CGG

C. Proposed dual modification model at CCGG



D. Lines of evidence for m4C/m5C competition

#	Evidence	Source	Strength
1	4mC at CCGG (Nanopore)	This study	Direct
2	5mC at GGCCGG (BS-seq)	Pisciotta 2023	Direct
3	4mC consensus = GGCCGG (100% of top 20 motifs)	This study	Strong
4	Dcm-like MTase +12x at T3 yet 4mC collapses 56%	This study	Strong
5	BS-seq cannot detect m4C (treated as unmethylated)	Methodology	Logical
6	Nanopore has low m5C sensitivity (<0.01%)	Methodology	Logical